SIMATS ENGINEERING
ASSESSMENT TOOL 1 — SOLUTIONS

COURSE NAME: ARTIFICIAL INTELLIGENCE
COURSE CODE: CSA17

ANALYTICAL PROBLEM SOLVING

Solution 1 — Water Jug Problem

1. Fill the 3-gallon jug completely. State: (4G = 0, 3G = 3)
2. Transfer all of it into the 4-gallon jug. State: (4G = 3, 3G = 0)
3. Fill the 3-gallon jug once more. State: (4G = 3, 3G = 3)
4. Pour from the 3-gallon jug into the 4-gallon jug until it is full (only 1 gallon fits), leaving 2 gallons behind. State: (4G = 4, 3G = 2)
5. Empty the 4-gallon jug onto the ground. State: (4G = 0, 3G = 2)
6. Pour the remaining 2 gallons into the 4-gallon jug. State: (4G = 2, 3G = 0)

Step	Action Taken	4-Gallon Jug	3-Gallon Jug
1	Fill the 3-gallon jug completely	0	3
2	Empty the 3-gallon jug into the 4-gallon jug	3	0
3	Fill the 3-gallon jug again	3	3
4	Top up the 4-gallon jug from the 3-gallon jug	4	2
5	Empty the 4-gallon jug onto the ground	0	2
6	Move the leftover 2 gallons into the 4-gallon jug	2	0

Result: 4-gallon jug = 2 gallons, 3-gallon jug = 0 gallons.

Solution 2 — Mars Rover as an Intelligent Agent

i. Percepts

- Camera imagery of the terrain
- Rock/soil composition readings
- Temperature
- Atmospheric pressure
- Wind speed and weather
- Obstacle-distance sensor data
- Terrain shape
- Battery/power status

ii. Environment Characteristics

1. Partially observable — sensors reveal only nearby surroundings.
2. Dynamic — weather and terrain can change unexpectedly.
3. Sequential — earlier actions affect later options.
4. Continuous — movement, distance and time vary smoothly.
5. Single-agent — the rover works without competing agents.

iii. Actions

- Move forward/backward, turn, stop
- Capture photographs
- Collect rock/soil samples

- Analyze samples
- Drill into rock
- Avoid obstacles
- Transmit data to Earth
- Recharge via solar panels

iv. Performance Evaluation

1. Scientific value of samples and data collected.
2. Distance travelled / area explored.
3. Success of data transmission to Earth.
4. Efficient use of battery and power.
5. Safety — avoiding damage or obstacles.

v. Suitable Agent Architecture

Utility-Based Agent — because the rover must weigh multiple possible actions against scientific value, safety, power cost, and mission priorities, picking the option with the greatest overall benefit.

Conclusion: The rover senses its environment, reasons over percepts, and acts through movement, sampling and communication in a partially observable, dynamic, sequential, continuous, single-agent world — best served by a utility-based architecture.

Solution 3 — The 8-Queens Problem

1. Initial State:

An empty 8×8 board with no queens placed.

2. State Space:

Every possible arrangement of queens across the board, based on which squares are occupied.

3. Actions:

- Place a queen on a safe square in the next column.
- Reposition a queen if the current placement causes a conflict.

4. Goal Test:

All 8 queens placed with none sharing a row, column, or diagonal.

5. Path Cost:

Each placement costs 1; since every solution needs 8 placements, path cost is not a distinguishing factor — any valid arrangement is acceptable.

Search Strategy: Backtracking

1. Place a queen in the first column.
2. Move to the next column and place a queen in a safe row.
3. Continue until all 8 queens are placed.
4. If no safe row exists, backtrack and try a different position for the previous queen.
5. Repeat until a solution is found.

One Valid Solution:

Column	Row
1	1
2	5
3	8
4	6
5	3

6	7
7	2
8	4

Board (Q = queen):

Q . . . .

. . . . . . Q .

. . . . Q . . .

. . . . Q

. Q . . . . . .

. . . Q . . . .

. . . . . Q . .

. . Q . . . . .

No two queens attack each other in this arrangement.

Solution 4 — OLA Cab Booking Agent

Agent Type: Goal-Based Agent

The customer's goal is simply to reach the destination; the app evaluates cab options and selects/executes the sequence of actions (enter destination, compare cabs, book, ride) that fulfils that goal.

Problem Formulation

Initial State:

- App open, current location detected.

Actions:

- Enter destination
- View available cabs
- Compare fare/comfort
- Select cab type
- Confirm booking

Goal State:

- Customer reaches the destination.

Path Cost:

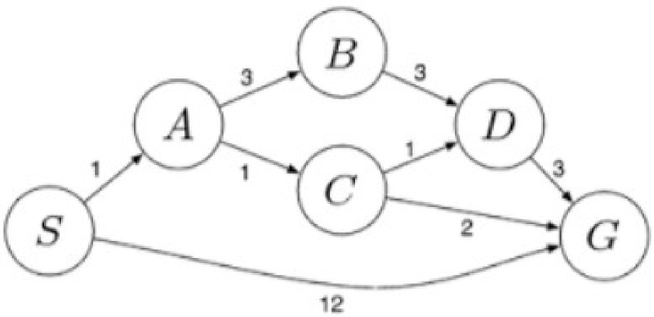
- Fare, travel time, waiting time.

Pseudocode

```
Algorithm OLA_CAB_BOOKING
Start
  1. Open OLA Application
  2. Detect Current Location
  3. Input Destination
  4. Display Available Cab Types
       Micro
       Mini
       Sedan
       Prime
       Shared
  5. Customer Selects Preferred Cab
```

```
   6. Check Cab Availability
   7. IF Cab Available THEN
          Calculate Fare
          Confirm Booking
          Assign Driver
          Start Trip
          Reach Destination
          Display "Trip Completed Successfully"
      ELSE
          Display "Selected Cab Not Available"
          Prompt Customer to Choose Another Cab
      END IF
Stop
```

Solution 5 — Least-Cost Routing with Uniform Cost Search



Step	Node Expanded	Cost	New Nodes Added
1	S	0	A(1), G(12)
2	A	1	B(4), C(2)
3	C	2	D(3), G(4)
4	D	3	G(6)
5	G	4	Goal Reached

Path	Total Cost
S → G	12
S → A → B → D → G	10
S → A → C → D → G	6
S → A → C → G	4

Least-Cost Path: S → A → C → G, Total Cost = 4 (1 + 1 + 2).